DATE- 18|01|25

UNIT - 1

SETS

* **Set**:- It is a Collection of well defined and distinct Objects. These objects are called the Element. and sets are represented in Curly Braces { } by a Capital letter.

Example:- $A = \{1, 2, 3, 4\}$

Set Name — Elements.

* **Representation of Sets**:-

(i) **Roster form**:- In this, we list all the elements of the set, separate them by 'Commas' and put them within 'Curly brackets' { }. It is also called 'Tabular form'.

Example:- (i) A set of Vowels $A = \{a, e, i, o, u\}$,

(ii) A set of natural number less than 7.

$B = \{1, 2, 3, 4, 5, 6\}$

(ii) **Set Builder form**:- In this, we do not list all the members or Elements of the set, but a rule is stated which any object must specify in order to be a member or Element of the set.

Example:- (i) All the natural numbers of the set A

$A = \{x : x \text{ is a natural number}\}$

(ii) $B = \{3, 4, 5, 6, 7, 8\}$ - Roster form

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$B = \{x \mid 3 \leq x \leq 8 \text{ and } x \in N\}$ - Set-Builder form.

* Types of Sets:-

(i) Null Set:- A set of no elements is called null set or Empty set. It is denoted by 'ϕ'.

Example:- $A = \{\ \}$ or $\{\phi\}$.

(ii) Unit Set:- A set which contains only one element is called 'unit set' or 'singleton set'

Example:- $A = \{5\}$,

(iii) Finite Set:- A set which contains finite number of elements is 'finite set'.

Example:- $A = \{a, b, c, d\}$ or $B = \{2, 4, 6, 8\}$

(iv) Infinite Set:- A set which contains an infinite number of elements is called an 'Infinite set'

Example :- $A = \{x : x$ is a natural number $\}$

(v) Universal Set :- A set which contains all the elements of different sets in a given problem is called 'Universal Set'. It is denoted by 'U'

Example :- If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$

$U = \{1, 2, 3, 4, 5, 6\}$

(vi) Subset :- Let A and B are two sets, then 'A' is said to be subset of 'B' If all the elements of set 'A' occur in set 'B'.

Example :- $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5, 6\}$ then '$A \subseteq B$'

(vii) Proper Subset : Let A & B are two sets, then set 'A' is said to be proper subset of 'B', If all the elements of set 'A' occur in set 'B' and set B contains atleast one more element which is not in set 'A'.

Example :- $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5\}$ then $A \subset B$.

viii) Number of Subset :- If any set 'A' contains 'n' elements then the number of its subset will be equal to 2^n.

Tatal number of Subsets = 2^n

Example :- $A = \{1, 2, 3\}$, no of Subsets = $2^3 = 8$

$\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\},$
$\{1, 3\}, \{1, 2, 3\}$.

(IX) Equal Sets : The two sets A and B are said to be equal if they have exactly the same elements, the order of Elements do not matter.

Example :- $A = \{1, 2, 3, 4\}$ and $B = \{4, 3, 2, 1\}$ then

$A = B$.

(X) Disjoint Sets :- The two sets A & B are said to be disjoint if the set does not contain any common element.

Example :- $A = \{1, 2, 3, 4\}$ & $B = \{5, 6, 7, 8\}$